# CODEALPHA INTERNSHIP PROGRAM

## INTERNET OF THINGS (IoT)

# TASK 2: SENSOR-BASED SIMULATION

## SMART STREET LIGHT AUTOMATION USING LDR SENSOR


**Submitted By**

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# CERTIFICATE OF COMPLETION

This is to certify that the project report entitled **"Smart Street Light Automation Using LDR Sensor"** has been successfully completed as part of the CodeAlpha IoT Internship Program.

The project demonstrates the implementation of an automated street lighting system using an LDR sensor and Arduino UNO through simulation. The system automatically controls street lights based on ambient light intensity, reducing energy consumption and improving efficiency.

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# ABSTRACT


Street lighting plays a vital role in ensuring safety and visibility during nighttime. Conventional street lighting systems often require manual operation and consume unnecessary electrical energy when left switched on during daylight hours.

This project presents a Smart Street Light Automation System using an LDR (Light Dependent Resistor) sensor and Arduino UNO. The system automatically detects surrounding light intensity and controls the street light accordingly. During daylight, the street light remains OFF, while during darkness it turns ON automatically.

The project demonstrates how IoT-based automation can improve energy efficiency, reduce operational costs, and contribute to smart city infrastructure development.


# INTRODUCTION

With increasing energy demands and urban development, efficient energy utilization has become a major concern. Smart lighting systems provide an effective solution by automating the operation of street lights.

An LDR sensor is capable of sensing ambient light levels and converting them into electrical signals. By interfacing the sensor with an Arduino UNO, the lighting system can automatically respond to environmental conditions.

This project simulates an intelligent street lighting system capable of operating without human intervention.

# OBJECTIVES

- To design an automatic street light control system.
- To monitor ambient light intensity using an LDR sensor.
- To reduce electrical power wastage.
- To demonstrate sensor-based automation using Arduino.
- To understand practical applications of IoT in smart infrastructure.

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# COMPONENTS USED

| Component | Quantity |
|---|---|
| Arduino UNO | 1 |
| LDR Sensor | 1 |
| LED | 1 |
| 220Ω Resistor | 1 |
| Breadboard | 1 |
| Jumper Wires | As Required |

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# CIRCUIT DIAGRAM

### Simulation Circuit

Figure 1: Complete Circuit Diagram of Smart Street Light Automation System.

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# WORKING PRINCIPLE

The Light Dependent Resistor (LDR) changes its resistance according to the intensity of light falling on it.

### Daytime Condition

- High light intensity is detected.
- Arduino receives a high analog value.
- LED remains OFF.
- Energy consumption is minimized.

### Nighttime Condition

- Light intensity decreases.
- Arduino detects darkness.

- LED turns ON automatically.
- Street illumination is provided.

The entire process occurs automatically without requiring human intervention.

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# ARDUINO PROGRAM

```
int ldr =
A0; int led
= 13;

void setup()
{
  pinMode(led, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  int value = analogRead(ldr);

  Serial.println(value);

  if(value < 500)
  {
    digitalWrite(led, HIGH);
  }
  else
  {
    digitalWrite(led, LOW);
  }

  delay(100);
}
```

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# CODE EXPLANATION

### Sensor Reading

```
analogRead(ldr);
```

Reads the light intensity from the LDR sensor.

### Decision Making

```
if(value < 500)
```

Checks whether the surrounding environment is dark.

### LED ON

```
digitalWrite(led, HIGH);
```

Turns ON the street light during darkness.

### LED OFF

```
digitalWrite(led, LOW);
```

Turns OFF the street light during daylight.

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# SIMULATION RESULTS

### Screenshot 1 – Complete Circuit

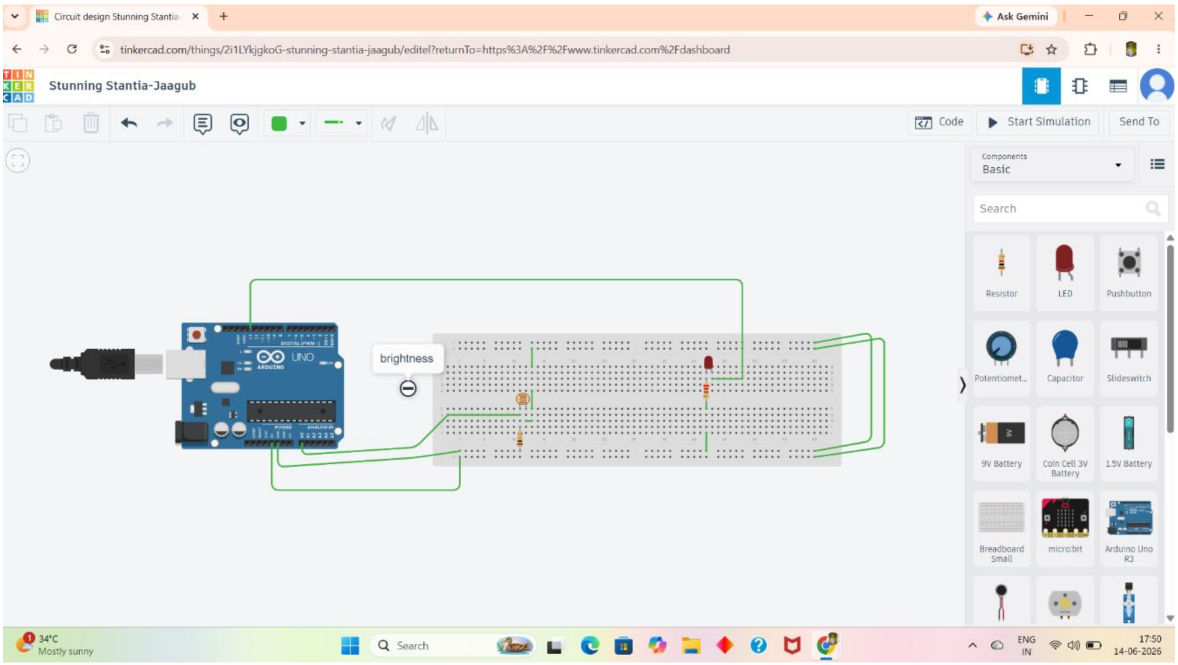


Figure 2: Complete Tinkercad Circuit.

## Screenshot 2 – Day Mode

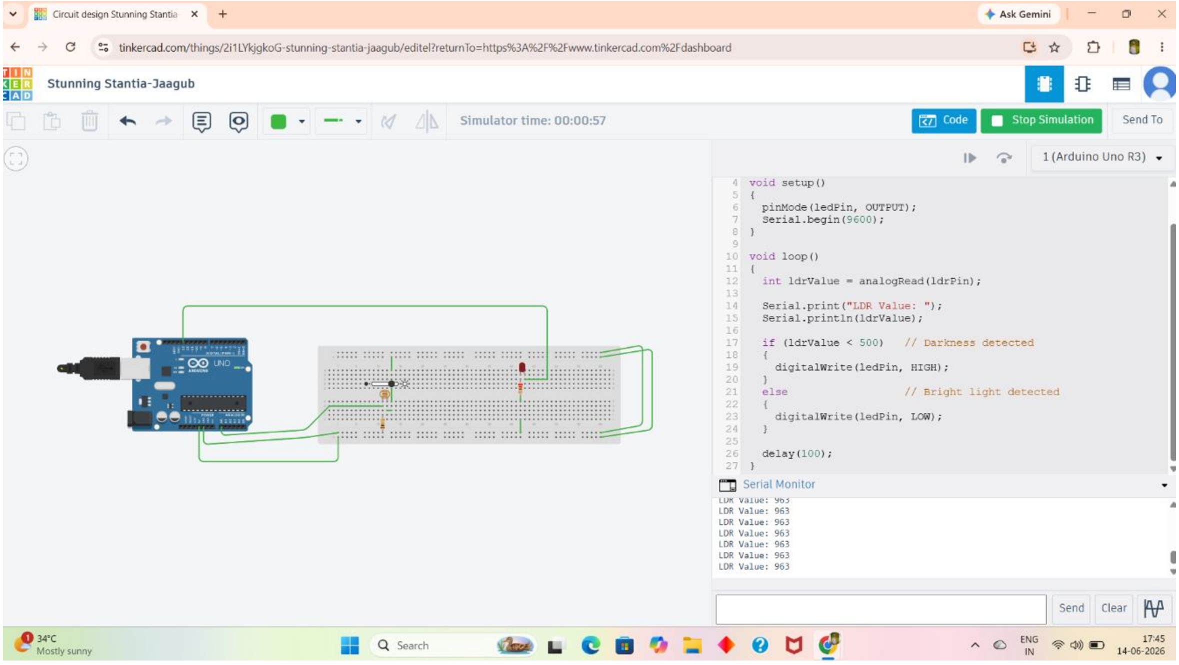


Figure 3: LED remains OFF during daylight conditions.

## Screenshot 3 – Night Mode

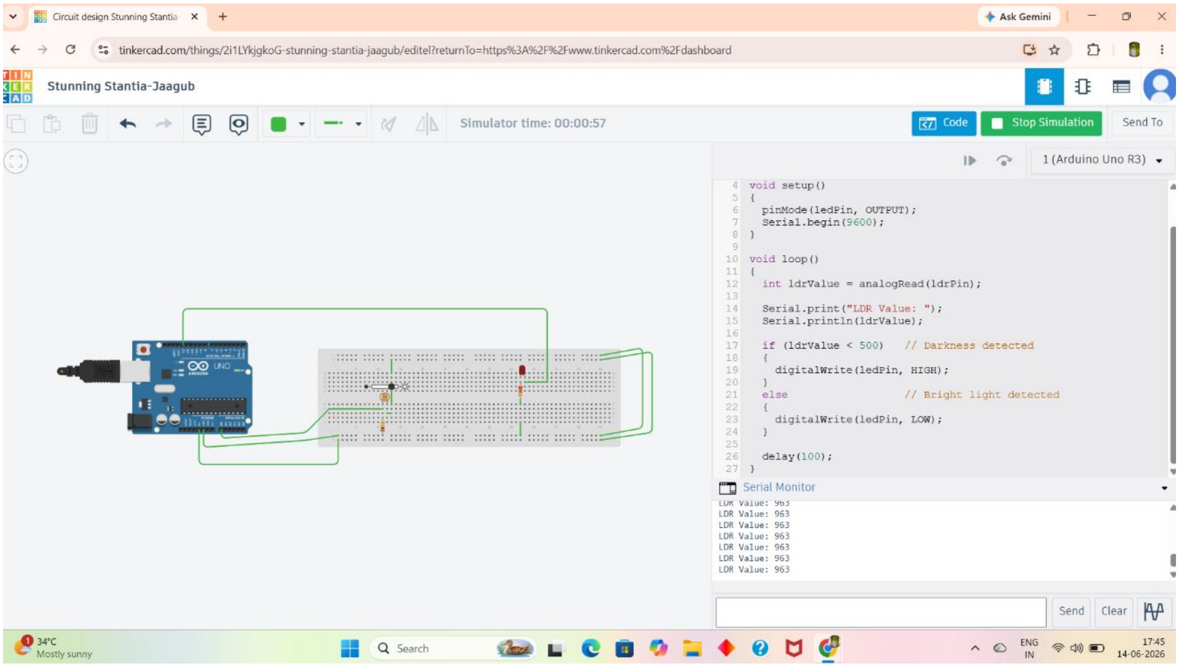


Figure 4: LED turns ON during darkness.

# APPLICATIONS

- Smart Street Lighting
- Smart City Infrastructure
- Parking Area Lighting
- Garden Lighting Systems
- Highway Lighting Control
- Campus Lighting Automation
- Industrial Outdoor Lighting

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# ADVANTAGES

- Energy Efficient
- Low Cost Implementation
- Automatic Operation
- Reduced Human Intervention
- Easy Installation
- Low Maintenance
- Environment Friendly
- Suitable for Smart Cities

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# FUTURE SCOPE

Future enhancements can include:

- Wi-Fi Based IoT Monitoring
- Cloud Data Storage
- Mobile Application Control
- Solar Powered Street Lighting
- Smart Brightness Adjustment
- Integration with Smart City Networks
- AI-Based Energy Optimization

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# RESULT

The Smart Street Light Automation System was successfully simulated using Arduino UNO and an LDR sensor.

The system accurately detected ambient light intensity and automatically controlled the LED based on environmental conditions. The simulation verified proper operation during both daytime and nighttime scenarios.

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# CONCLUSION

The Smart Street Light Automation System demonstrates a simple and effective IoT-based solution for automatic lighting control. By using an LDR sensor and Arduino UNO, the system reduces energy wastage and eliminates the need for manual operation.

The project highlights the practical implementation of sensor-based automation and its importance in developing energy-efficient smart city infrastructure.

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# REFERENCES


1. Arduino UNO Official Documentation
2. Tinkercad Circuits Simulation Platform
3. LDR Sensor Datasheet
4. IoT Fundamentals and Applications
5. Smart City Infrastructure Research Papers


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# ACKNOWLEDGEMENT


I express my sincere gratitude to CodeAlpha for providing the opportunity to work on this project under the Internet of Things Internship Program. This project enhanced my understanding of Arduino programming, sensor interfacing, automation systems, and IoT applications in real-world scenarios.